

# Disentangling anthropogenic stimuli: Visual and acoustic components differentially impact captive pied tamarin behaviour

Fiene Steinbrecher<sup>c,a,\*</sup>, Eluned C. Price<sup>b</sup>, Claudia A.F. Wascher<sup>c</sup>, Fay E. Clark<sup>c,d</sup>, Jacob C. Dunn<sup>c,e,f,\*\*</sup>

<sup>a</sup> Durrell Wildlife Conservation Trust, Jersey, British Channel Islands, United Kingdom

<sup>b</sup> Tamarin Trust, Jersey, British Channel Islands, United Kingdom

<sup>c</sup> Behavioural Ecology Research Group, School of Life Sciences, Anglia Ruskin University, United Kingdom

<sup>d</sup> School of Psychological Sciences, University of Bristol, United Kingdom

<sup>e</sup> Department of Behavioural and Cognitive Biology, University of Vienna, United Kingdom

<sup>f</sup> ENES Bioacoustics Research Laboratory, University of Saint-Étienne, Saint-Étienne 42023, France

## ARTICLE INFO

### Keywords:

Animal behaviour  
Multimodal  
Playback  
Callitrichid  
Welfare  
Bioacoustics

## ABSTRACT

Anthropogenic activity is increasingly disturbing wildlife, leading to negative impacts on species welfare and conservation. Despite this, very few studies have investigated how the different components of anthropogenic activity (including visual, acoustic, or other components) differentially affect animals. Such studies are important to better understand what animals may perceive as aversive in order to improve, not only captive management and welfare, but also wild management. This study aimed to disentangle the impacts of the visual and acoustic components of two different anthropogenic stimuli (zoo visitors and construction activity) on the critically endangered pied tamarin (*Saguinus bicolor*). We exposed 12 pair-housed pied tamarins at Jersey Zoo to either no stimuli (baseline), individual components (visual or acoustic stimuli only), or combinations of components (54 trials in total). We found that pied tamarins varied in their responses to the two different stimuli and to their different components, especially when presented in isolation. Visitor visual and visitor acoustic components presented in isolation led to increased vigilance. The combined presentation of visitor components weakened responses, indicating that simultaneous exposure to visual and auditory cues may provide more perceptual context to the primates. In contrast, construction noise caused a significant decrease in environmental engagement (indicated by decreased foraging and increased retreat behaviour). These findings emphasize the importance of understanding how animals respond to different sensory components of stimuli to improve captive management practices, enhance welfare, and ultimately help the conservation of endangered species.

## 1. Introduction

Animals in zoos inhabit environments that, while designed for their care, are inherently artificial and shaped extensively by human activity. As a result, human-generated sights, sounds, and smells, ranging from visitor presence to construction or routine maintenance, become embedded in the sensory landscape that animals must navigate. Although numerous studies have investigated the effects of anthropogenic disturbance on free-living wildlife (Harding et al., 2019; Shannon et al., 2016), very few studies have studied how the specific sensory components of these anthropogenic activities, such as visual (e.g. human

presence, movement, and novel objects) and acoustic (e.g. human voices and mechanical noise) components, differentially affect free-living and captive wildlife (Clark and Dunn, 2022; Halfwerk and Slabbekoorn, 2015; Munoz and Blumstein, 2012). This leaves a significant gap in the research. The majority of studies have focused on the overall effect of human stimuli, typically characterized as the ‘visitor effect’ (Collins et al., 2017; Davey, 2007; Farrand, 2007; Fernandez et al., 2009; Hosey, 2000; Sherwen and Hemsworth, 2019; Suárez et al., 2017), rather than determining which components of human activity drive behavioural or welfare responses. Only by separating and testing an animal’s response in these ecologically relevant sensory components, both in isolation and

\* Correspondence to: Durrell Wildlife Conservation Trust, Les Augrès Manor, La Profonde Tue, Trinity, JE3 5BP, Jersey, British Channel Islands, United Kingdom.

\*\* Correspondence to: Anglia Ruskin University, East Road, Cambridge CB1 1PT, United Kingdom.

E-mail addresses: [fiene.steinbrecher@gmail.com](mailto:fiene.steinbrecher@gmail.com) (F. Steinbrecher), [jacob.dunn@aru.ac.uk](mailto:jacob.dunn@aru.ac.uk) (J.C. Dunn).

<https://doi.org/10.1016/j.applanim.2026.106926>

Received 30 September 2025; Received in revised form 10 January 2026; Accepted 18 January 2026

Available online 19 January 2026

0168-1591/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

in combination, can we fully understand the ways in which they disturb them and make better evidence-based animal management decisions.

The current study used pied tamarins (*Saguinus bicolor*) as a model species that is threatened with extinction in the wild, acoustically sensitive, and important to manage effectively in *ex-situ* populations (zoos). Anthropogenic activities pose a pervasive and ongoing challenge for pied tamarins, affecting them both in their natural habitats and in captivity (Gordo et al., 2021; Sobroza et al., 2024a, 2024b; Steinbrecher et al., 2023; Wormell et al., 1996). In the wild, pied tamarins inhabit forest patches that are increasingly fragmented, and rapidly being lost, in and around Manaus, Brazil, which is one of the fastest growing cities in the Amazon Basin (Gordo et al., 2013). Pied tamarins have the smallest range of any Amazonian primate (~7500 km<sup>2</sup>) (Gordo et al., 2013), are classified as Critically Endangered by the IUCN (Gordo et al., 2021), and are considered one of the world's 25 most endangered primate species (Mittermeier et al., 2024). Captive pied tamarins often show highly activated physiological and behavioural stress responses (Price et al., 2019), and have chronic physical health problems (Cabana et al., 2018; Richards-Rios et al., 2021) and poor breeding success (Armstrong and Santymire, 2013; López et al., 2001; Wormell et al., 1996).

We examined pied tamarin behaviour and use of space in response to the visual and acoustic components of two anthropogenic activities: 1) close enclosure proximity of zoo visitors; and 2) construction activity. Based on the threat-sensitive predator avoidance hypothesis (Bishop and Brown, 1992; Lind and Cresswell, 2005; Monclús et al., 2009; Papworth et al., 2013), we expected pied tamarins to evaluate and respond to anthropogenic activity according to whether they perceived it as threatening or not. This would be shown in their behaviour and use of space. We predicted that if stimuli were perceived as threatening, pied tamarins would move away and position themselves higher in the enclosure and/or increase defensive or avoidance behaviours, but that no changes would be observable if stimuli were perceived as non-threatening. We also hypothesised that congruent conditions (i.e., two components from the same stimulus, e.g., acoustic and visual components of visitors) would provide more relevant information to tamarins, and therefore cause a different response than incongruent conditions or components in isolation. More specifically we predicted that when confronted with incongruent stimuli (i.e., two components from different stimuli, e.g., acoustic component of construction and visual components of visitors) or components in isolation, the information would be less reliable for the tamarins to assess a potential threat, and thus they would attempt to gather more information, and/or positioning themselves higher to increase visibility and safety.

## 2. Methods

### 2.1. Subjects and housing

Experiments were conducted on 12 pair-housed pied tamarins housed at Jersey Zoo, British Channel Islands, between May and August 2022. Subjects were five males and seven females, ranging from seven to 23 years of age. All subjects were captive bred and kept in non-breeding pairs; three of these were mixed-sex and three were same-sex (Table S1).

Pairs of tamarins were housed in similar enclosures in three different building locations across the zoo. Enclosures had indoor (mean 18 m<sup>3</sup>, SD = 8.19) and an outdoor (mean 68 m<sup>3</sup>, SD = 38.57) areas connected by a small (0.5 × 0.5 m) opening with a sliding door. Pied tamarins housed in the same building were separated by an empty enclosure between them, thus avoiding any direct contact. Subjects had access to their outdoor areas during the day (08:00–16:30 h), but were kept indoors overnight. The indoor areas were equipped with a nest box, feeding platform, and several wooden planks, ladders, and ropes. The outdoor areas had a variety of plants and foliage and were equipped with tree trunks, ropes, and natural branches. Subjects were fed a mix of commercial primate pellets in the morning (08:30–09:30 h), fruits and

vegetables at noon (12:00–13:00 h), and live insects in the afternoon (16:00–17:00 h). Food was provided in bowls and live insects were scattered on their food platforms.

### 2.2. Experimental design

The experiment consisted of a 3 × 3 paradigm with nine conditions: a baseline (no stimuli); two visual stimuli (the visual component of visitors or construction); two acoustic stimuli (the acoustic component of visitors or construction); and four combined visual and acoustic stimuli that were either congruent (e.g., the visual and acoustic components of visitors) or incongruent (e.g., the visual component of visitors and the acoustic component of construction) (Table 1).

### 2.3. Experimental procedure

Each tamarin pair was exposed to each of the nine experimental conditions (Table 1) once, yielding 54 trials. Each tamarin pair participated in one trial per day. The order of conditions for each pair was randomised without replacement using a random number generator, although if one of the pairs was exposed to sound playback, the other pair at the same location was not exposed to sound playback on the same day (i.e., pairs 2, 3 and 4 or pairs 5 and 6; Table S1). Trials took place between 13:00 and 16:00 h to avoid coinciding with routine husbandry, and each trial was 5-min duration with approximately 3-min setup and 3-min take-down time either side. For all conditions including baseline, the speaker was installed within 3-min of arrival and the observer moved in front of the enclosure to allow enough time for the subjects to settle and behave naturally. A trial was only started when both tamarins were in their outside enclosure. For conditions with a visual and/or acoustic component, the visual component was put in place, and/or acoustic playback started, immediately before the start of the 5-min trial. At the end of the 5-min trial, all stimuli were removed, and the researchers left within 3-min.

### 2.4. Visual stimuli

**Visitor visual stimulus:** Three adult zoo volunteers behaved as a cohesive group. They were dressed in ordinary, day-to-day clothes (not resembling zoo staff, who wear uniforms). All volunteers were unfamiliar to the subjects. Although the volunteers were not real zoo visitors, their unfamiliarity and lack of prior contact with the subjects, as well as their visitor-like behaviour and appearance, ensured they were likely to be perceived similarly, allowing us to treat them as “visitors” in this context of this experiment. During visitor visual conditions, the group walked slowly towards the outside enclosure until they were 1 m in front of it, then stood in silence. **Construction visual stimulus:** A mock construction site was installed at the same distance as the visitor visual stimulus, 1 m in front of the enclosure. This consisted of two red and white plastic traffic cones (~71 cm height) spaced 1.5 m apart with two rows of chains hanging between. A bright yellow and black construction sign was placed next to one of the cones (~50 cm height). This combination of items meant that installation was quick at the start of a trial. In addition, the construction mock-up site was intentionally designed without human workers, featuring only stationary objects typically found at construction sites, to make a clear distinction between the visual components of the two stimuli.

### 2.5. Acoustic stimuli

**Visitor acoustic stimulus:** The sounds of natural human conversation including spontaneous laughter and brief moments of silence was recorded from five adult zoo volunteers and edited to match the acoustic characteristics of typical visitor noise. **Construction acoustic stimulus:** The sounds of hammering, drilling, metal cutting and ‘clanging’ of metal pieces was recorded from real construction work taking place

**Table 1**

3×3 matrix showing the nine experimental conditions of the multimodal experiment of visual and acoustic stimuli. Conditions were grouped into visual (blue), acoustic (orange), congruent (green) and incongruent (purple).

|                  |                    | Visual stimuli     |                                     |                                          |
|------------------|--------------------|--------------------|-------------------------------------|------------------------------------------|
|                  |                    | No visual          | Visitor visual                      | Construction visual                      |
| Acoustic stimuli | No sound           | Baseline           | Visitor visual                      | Construction visual                      |
|                  | Visitor sound      | Visitor sound      | Visitor sound + Visitor visual      | Visitor sound + Construction visual      |
|                  | Construction sound | Construction sound | Construction sound + Visitor visual | Construction sound + Construction visual |

approximately 150 m from the closest study enclosures. Recordings were made at a 1–2 m horizontal distance away from the epicenter of the source at a sampling rate of 44.1 kHz with a 16-bit resolution, using a linear PCM Tascam DR40 recorder (TEAC Corporation, Tokyo, Japan) and a handheld Shotgun Microphone (Rode, Silverwater, Australia) with a windshield. The sound files were then edited with Audacity software (Audacity® version 3.0.2) to produce a 5-min file (.wav file) for each stimulus, which included a 30-sec fading-in period at the beginning of the recording and a 30-sec fading-out period at the end to avoid startling the animals with an abrupt change (Wale et al., 2013).

Playback files were played through a speaker (Chiayo Focus Pro U0PC) positioned on the ground 1.5 m from the outdoor enclosure and oriented towards the center of it. The speaker was camouflaged with vegetation and playback was started remotely via Bluetooth. To control for the possibility of the speaker also being a visual stimulus, it was also placed in the same location (but switched off) for all other conditions including baseline. The sound pressure levels of playbacks were based on the sound pressure levels (measured with an SPL meter (Precision Gold NO5CC Sound Level Meter, Maplin Electronics Ltd, Rotherham, UK)) at which they ‘naturally’ occurred when the recordings were made (visitor sound: peak amplitude 65 dB SPL at 1.5 m; construction sound: peak amplitude 75 dB SPL at a 1.5 m distance).

## 2.6. Data collection

Observations of animals were only made when subjects were present in their outdoor enclosures. Behavioural data were collected using a combination of instantaneous sampling and one-zero sampling, as different behaviours measures required different recording approaches (Altmann, 1974). All behaviours were recorded referring to an ethogram (Table 2).

Instantaneous sampling was used to record state variables that could be reliably assessed at a single moment in time (Altmann, 1974), including the individual’s general activity state and their use of space within the enclosure. The subject’s use of space in the enclosure was recorded in height and depth categories. For this, the enclosure was visually divided into three equal height levels, low (including the ground), middle, and high; and three equal depth levels, front (including the enclosure fence) middle, and back (Figure S1).

One-zero sampling was used to record the occurrence of discrete or infrequent event behaviours (i.e., looking, chatting, scratching, and scent marking), where the primary interest was whether a behaviour occurred within a defined time window rather than its frequency (Bassett et al., 2003; Blumstein et al., 2006).

**Table 2**

Ethogram of state and event behaviours.

| Behaviour               | Description                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>State behaviours</i> |                                                                                                                                                                       |
| Foraging                | Actively searching for or manipulating the environment to locate, extract, or obtain food.                                                                            |
| Feeding                 | Actively drinking or consuming a food item that was provided or found by the subject.                                                                                 |
| Vigilance               | Scanning the surroundings with frequent, swift head movements, indicating an alert state.                                                                             |
| Moving                  | Locomoting within the enclosure, such as walking, running, or climbing.                                                                                               |
| Resting                 | Remaining stationary and non-alert, often lying down with a curled tail; may be asleep or awake.                                                                      |
| Retreating              | Staying out of view in the indoor enclosure or other secluded area.                                                                                                   |
| Grooming                | Engaging in fur maintenance behaviours; often deliberate and controlled movements to search through or clean their own fur.                                           |
| NA                      | Subjects were out of sight. This was only recorded when subjects were known to be in the outdoor enclosure, but unable to be located at the time of the interval.     |
| <i>Event behaviours</i> |                                                                                                                                                                       |
| Looking                 | Directly gazing towards the stimulus, typically towards the front of the outdoor enclosure where the stimulus was placed or played from.                              |
| Chattering              | Producing a vocalisation associated with a threat display (Sobroza et al., 2017), often accompanied by an alert gaze and side-to-side head movements.                 |
| Scratching              | Rapid and repetitive movements using hands or feet to scratch the body.                                                                                               |
| Scent marking           | Rubbing different parts of the body (e.g., genital, sub-caudal, suprapubic, sternal, and head) against objects in the enclosure to communicate and/or mark territory. |

Sampling was conducted at 15-second intervals. Within each pair, observations alternated between individuals, resulting in a 30-second sampling interval per individual and a total of 10 sampling points per individual per 5-min trial. This alternating approach was necessary due to dense foliage within the enclosures, which limited continuous simultaneous visibility of both subjects and therefore allowed for more accurate and focused recording of each individual’s behaviour.

## 2.7. Statistical analyses

All analyses were performed using R version 3.5.2 (R Core Team, 2022). A Principal Component Analysis (PCA) was conducted to reduce the number of observed behaviours into fewer variables and therefore minimise model complexity. As several behaviours were assumed to be correlated, PCA was carried out with an oblique rotation (i.e., Promax

rotation), which allows for components to be correlate. This approach allowed us to consider and identify these correlations, reduce the number of variables, simplify models for easier interpretability, and reduce statistical errors, particular for an experiment with relatively small sample size (Budaev, 2010; Costello and Osborne, 2019). PCA analysis included all frequently observed behaviours (forage, vigilant, move, rest, retreat, look, chatter and scratch) from all trials and was done using the 'prcomp' function from the 'FactoMineR' package (Lé et al., 2008) as well as the 'psych::principal' function from the 'psych' package (Revelle, 2023). Other behaviours were excluded from the analysis, due to their limited number of occurrences (less than five occurrences). Principal Components (PCs) were included in further analyses based on inspection of the Kaiser criterion (inclusion of components with an eigenvalue greater than 1), the empirical Bayesian Information Criterion (BIC; models with the lowest BIC are preferred), the Very Simple Structure (VSS; evaluation of the best fitting individual components/factors) and Standardized Root Mean Square Residuals (or SRMR; average difference between the implied and the observed correlation) (Morton and Altschul, 2019).

The extracted PCs were analysed with generalised linear mixed-effects models (GLMM) using the lme4 (Bates et al., 2015) and glmmTMB packages (Brooks et al., 2017). Each PC was a dependent variable analysed in an individual model. To investigate the different effect of each condition on the PCs, each component (visual and acoustic) was classified by its three possible presentations (no stimulus, visitor stimulus, and construction stimulus) and used as an independent variable with three levels. To account for repeated observations, subject ID was included as a random factor in the model. Model assumptions were tested using the DHARMA package (Hartig, 2022), which provides simulation-based residual diagnostics to test for dispersion, zero-inflation, and uniformity. In addition, we visually examined QQ-plots to confirm adequate model fit. Assumptions were satisfied for PC1 and PC2, but not for PC3. Therefore, a log transformation was applied to the PC3 model, which resolved the violation of assumptions. The subjects' use of space was analysed following the same model structure and analysis.

## 2.8. Ethical note

This study was ethically approved by Durrell Wildlife Conservation Trust's Ethics Committee (#ETH19/1112, 21 January 2019) and Anglia Ruskin University (FREP/SREP number: A&EB DREP19-126, 09/12/19). All subjects were housed under their normal management conditions and were not deprived of food or water. Playbacks consisted of sounds that already occurred around the zoo and thus were not entirely novel to the animals. In addition, playbacks were played at noise amplitude levels that did not exceed normal levels of each stimulus.

Human speech recordings consisted of five zoo volunteers having a conversation. Prior to recordings, the participants were fully informed about the use of the recordings in the experiment and the experiment itself. All participants gave verbal consent and no personal or sensitive information was discussed nor recorded. In addition, recordings were deleted after the conclusion of the experiment.

## 3. Results

### 3.1. Principal component analysis (PCA)

Three Principal Components (PCs) were included for further analysis based on the criteria described in the methods. The first three components explained a combined 72.2 % of the total variance (PC1 = 35.5 %; PC2 = 19.9 %; and PC3 = 16.8 %).

PC1 had positive loadings for vigilance, looking, and chattering (Table 3). PC1 was thus characterised as 'vigilance'. In contrast, PC2 had positive loadings for foraging and negative loadings for retreating. We therefore labelled PC2 as 'environmental engagement'. PC3 had positive

**Table 3**

The eight behaviour scores grouped by Principal Component.

| Behaviour              | PC1<br>Vigilance | PC2<br>Environmental Engagement | PC3<br>Rest   |
|------------------------|------------------|---------------------------------|---------------|
| Foraging               | -0.0018          | <b>1.0056</b>                   | -0.0007       |
| Vigilance              | <b>0.9681</b>    | -0.0001                         | -0.0000       |
| Moving                 | 0.0000           | 0.0000                          | 0.0000        |
| Resting                | 0.0000           | -0.0000                         | <b>1.0000</b> |
| Retreating             | -0.0553          | <b>-0.3550</b>                  | -0.0157       |
| Looking                | <b>0.9221</b>    | 0.0000                          | 0.0004        |
| Chattering             | <b>0.2079</b>    | -0.0000                         | -0.0003       |
| Scratching             | 0.0000           | -0.0000                         | 0.0002        |
| Eigenvalue             | 2.87             | 1.59                            | 1.34          |
| Variance explained (%) | 35.9             | 19.9                            | 16.8          |

Variables in bold contribute the highest amount to each behaviour variable.

loadings for resting. Thus, PC3 was considered a general measure of 'rest' (Table 3).

### 3.2. Behavioural analysis

Compared to baseline, the behaviour of pied tamarins was found to be significantly affected by experimental condition.

PC1 ('vigilance') was significantly impacted by visual components (GLMM:  $F_{2,88} = 30.382$ ,  $P < 0.001$ ), and the interaction between visual and acoustic components (GLMM:  $F_{4,88} = 3.075$ ,  $P = 0.020$ ). These findings were driven by the significant positive impact of the visual component of visitors (GLMM:  $t_{88} = 5.605$ ,  $P < 0.001$ ) and the acoustic component of visitors (GLMM:  $t_{88} = 2.269$ ,  $P = 0.026$ ), and the significant negative interaction between the visual and acoustic components of visitors (GLMM:  $t_{88} = -2.925$ ,  $P = 0.005$ ). Given that vigilance, looking and chattering were all positively loaded onto PC1, significant positive effects meant an increase in these behaviours, whereas significant negative effects meant a decrease. No other condition had a significant effect on PC1 (Fig. 1a; Fig. 2a).

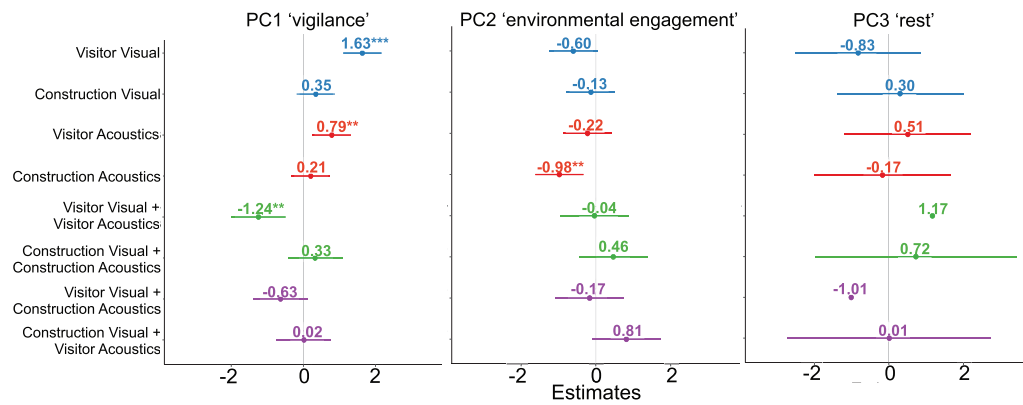
PC2 ('environmental engagement') was significantly affected by the acoustic components (ANOVA:  $F_{2,88} = 4.307$ ,  $P = 0.016$ ). This finding was driven by the significant negative effect of the acoustic components of construction on PC2 (GLMM:  $t_{88} = 2.988$ ,  $P = 0.041$ ). As foraging was positively loaded, and retreating negatively loaded onto PC2, significant negative effects on PC2 meant a decrease in foraging and an increase in retreating behaviour. No other condition had a significant impact on PC2 (Fig. 1b; Fig. 2b).

PC3 ('rest') was not predicted by experimental condition specifically or overall (Fig. 1c; Fig. 2c).

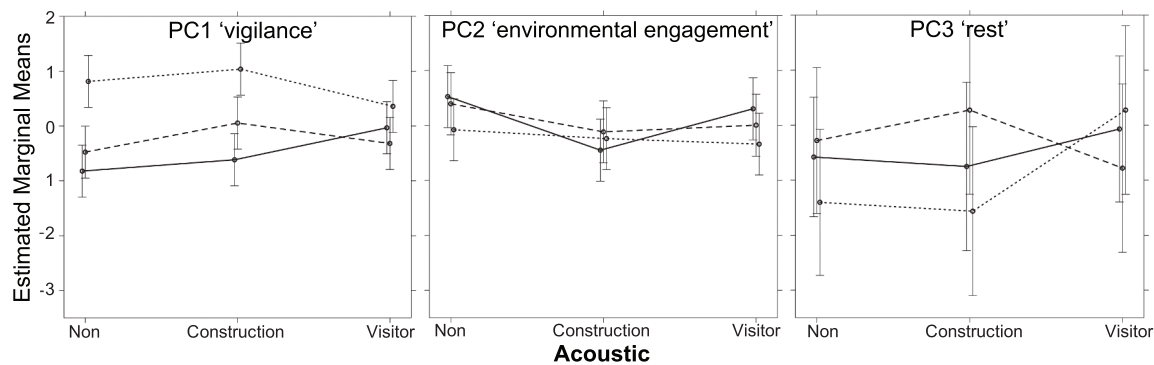
### 3.3. Use of space analysis

Use of space in the enclosure was analysed by both depth (front, middle, back) and height (low, middle, high). Depth in the enclosure was significantly influenced by visual components (GLMM:  $F_{2754} = 23.475$ ,  $P < 0.001$ ), acoustic components (GLMM:  $F_{2754} = 7.105$ ,  $P = 0.001$ ), and the interaction between them (GLMM:  $F_{2755} = 4.236$ ,  $P = 0.002$ ). This was largely driven by the strong positive impact (i.e., increase in depth) of the construction visual (GLMM:  $t_{754} = 2.205$ ,  $P = 0.028$ ) and the construction acoustic components (GLMM:  $t_{754} = 4.317$ ,  $P < 0.001$ ) as well as the interaction between the visual and acoustic components of construction (GLMM:  $t_{755} = -3.360$ ,  $P = 0.001$ ) and the visual component of visitor and the acoustic component of construction (GLMM:  $t_{755} = -2.989$ ,  $P = 0.003$ ) (Fig. 3a; Fig. 4a).

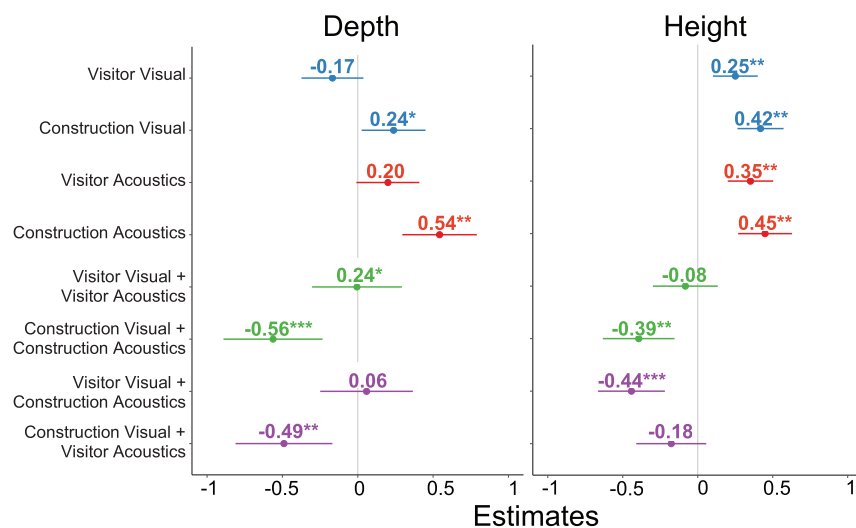
Height in the enclosure was significantly influenced by the visual components (GLMM:  $F_{2754} = 6.746$ ,  $P = 0.001$ ), the acoustic components (GLMM:  $F_{2754} = 15.873$ ,  $P < 0.001$ ) and the interaction between them (GLMM:  $F_{2755} = 4.958$ ,  $P = 0.001$ ). This was largely driven by the strong positive impact (i.e., increase in height) of the visual component of visitors (GLMM:  $t_{754} = 3.297$ ,  $P = 0.001$ ), the visual component of



**Fig. 1.** Coefficient estimate plots of the GLMMs for the three PCs. Experimental conditions are represented on the y-axis. The whiskers extending from each estimate indicate the 95 % confidence intervals. Conditions are grouped into visual stimuli (blue), acoustic stimuli (red), congruent stimuli (green), and incongruent stimuli (purple). The results show the effects of each condition compared to baseline, with the vertical line in each panel representing the null hypothesis of no significant difference from baseline. Values marked with asterisks indicate significant differences between baseline and the condition (\*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$ ).



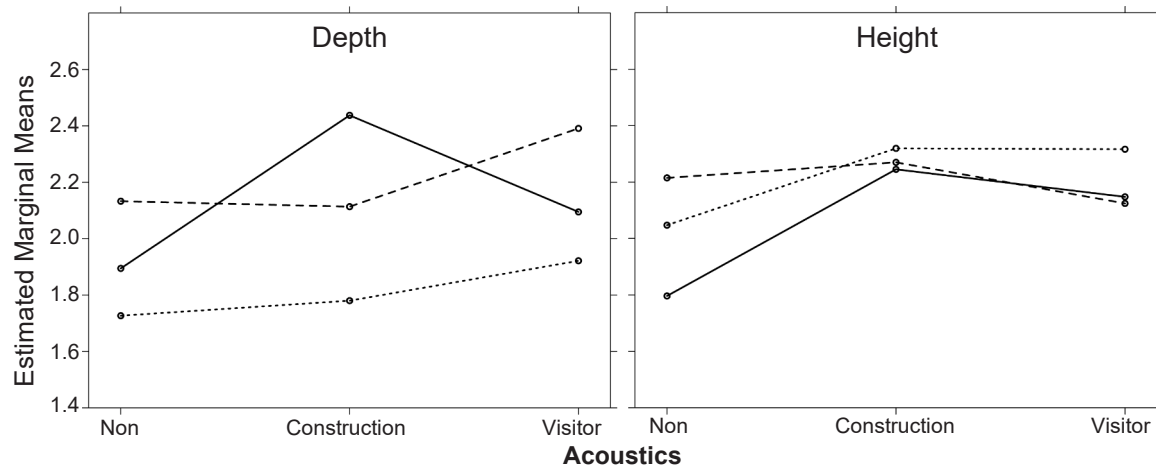
**Fig. 2.** Estimated marginal mean (EMM) plots for Principal Components 1–3. The acoustic components are categorised on the x-axis (non, construction, and visitor) and the line represents the visual component: no visual (solid line), construction visual (dashed line), and visitor visual (dotted line). Error bars indicate  $\pm 1$  standard error of the mean.



**Fig. 3.** Coefficient estimate plots of the GLMMs for use of space (depth and height). Experimental conditions are represented on the y-axis. The whiskers extending from each estimate indicate the 95 % confidence intervals. Conditions are grouped into visual stimuli (blue), acoustic stimuli (red), congruent stimuli (green), and incongruent stimuli (purple). The results show the effects of each condition compared to baseline, with the vertical line in each panel representing the null hypothesis of no significant difference from baseline. Values marked with asterisks indicate statistical significance between baseline and the condition (\*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$ ).

construction (GLMM:  $t_{754} = 5.335$ ,  $P < 0.001$ ), the acoustic component of visitor (GLMM:  $t_{754} = 4.549$ ,  $P < 0.001$ ) and the acoustic component





**Fig. 4.** EMM plots for use of enclosure space depth and height. The acoustic stimuli are categorised on the x-axis (non, construction, and visitor) and the line represents the visual stimuli (no visual (solid line), construction visual (dashed line), and visitor visual (dotted line).

of construction ( $t_{754} = 4.991$ ,  $P < 0.001$ ). Additionally, there was a negative interaction (i.e., decrease in height) between the visual and acoustic components of construction (GLMM:  $t_{755} = -3.328$ ,  $P = 0.001$ ) and between visual component of construction and the acoustic component of visitors (GLMM:  $t_{755} = -3.888$ ,  $P = 0.000$ ) (Fig. 3b; Fig. 4b).

#### 4. Discussion

This study examined the behavioural responses of pied tamarins, a critically endangered primate species, to the visual and acoustic components of zoo visitors and construction activity. Our results revealed four key findings: (1) different anthropogenic stimuli (visitors, construction) and the components comprising those anthropogenic stimuli (visual, acoustic), caused significant differences in behavioural responses and use of space; (2) isolated components (only visual or acoustic) generally elicited stronger behavioural responses than combinations of components; (3) incongruent combinations of components did not lead to any significant changes in behaviour, while congruous combinations of components for visitors showed a significantly lower response than isolated stimuli; (4) both visual and acoustic components influenced the use of space, although construction sound had a particularly pronounced effect on the depth that tamarins went to in the enclosure, while both types of anthropogenic stimuli (visitor, construction) had significant impacts on height.

##### 4.1. Responses to visitor stimuli

Pied tamarins, belonging to the family Callitrichidae (tamarins, lion tamarins, and marmosets), are among the smallest primate species, with an adult weight of around 500 g (Gordo et al., 2013). They face predation from a wide range of animals (Hart, 2007) and have developed diverse antipredation strategies aimed at early detection of potential threats (Ferrari, 2009; Ferrari and Ferrari, 1990). Therefore, when confronted with visitors, an increase in vigilance, as seen in our results, was expected. Looking more closely at visitors as a visual stimulus may explain why they elicit a strong response from pied tamarins. Visitors approaching the enclosure from different angles, often using unpredictable and sudden movements, gestures, and facial expressions, could potentially convey dominance, aggression, or a potential threat. For example, eye contact or teeth baring are well characterized behavioural indicators of aggression and dominance in nonhuman primates (Kim et al., 2022; Petersen et al., 2018), and thus visitors looking at or smiling could have been perceived as threatening, frightening, or stressful (Chiew et al., 2019; Hosey, 2000, 2013). The use of space analysis

corroborated the strong significant effect of visitors on the ‘vigilance’ response when tamarins were confronted with visitors visually. Although statistically insignificant, pied tamarins chose to be closer to the front of the enclosure in the presence of visitors, compared to conditions with no visitor presence or the construction visual condition. Similar results were shown by Wormell et al. (1996), who also found that zoo-housed pied tamarins responded to visitors with approach and threat (piloerection and chatter) more often than other callitrichid species.

Moreover, the unfamiliarity of the visitors could have reinforced increased vigilance responses. Although all of our study subjects lived in a zoo and were familiar with the presence of visitors, most of them were housed in off-show enclosures inaccessible to visitors. This suggested that while they might have been housed in on-show enclosures at some point and thus had experienced visitors before, visitors were still a novelty. Past research has shown that zoo-housed primates can differentiate between different categories of human, such as keepers, staff and unfamiliar visitors (Smith, 2014, 2016) and therefore familiarity (i.e., habituation) played an important role. For practicality reasons, the inclusion of more on-show individuals in our study sample was not possible and therefore, an extension or repetition of the study, including on-show individuals, might provide more insight into this possibility. However, for tamarins that are on-show, visitor noise and visitor behaviour near a test group would be harder to control.

##### 4.2. Responses to construction stimuli

Construction sound in isolation caused a decrease in environmental engagement (PC2). In addition, construction sound in isolation was associated with changes in enclosure space use. Pied tamarins either retreated to the indoor enclosure or preferred to stay at the back of the outdoor enclosure when confronted with the acoustic component alone (depth). One possible explanation may be the design of the construction stimulus. For this experiment, the construction mock-up site was intentionally designed not to include people and only feature stationary objects typically found at construction sites, to make a clear distinction from the visual components of the visitor stimulus. This possibly made the construction stimulus a rather static and perceptually dull event against the normal zoo background. In contrast, the acoustic construction stimulus could have been a perceptually strong event, with the peak amplitude of the construction playback equivalent to the lower end of construction noise levels of 75 dB (Neitzel et al., 1999). This was in fact 10 dB higher than the level produced by visitors, and therefore twice as loud. Furthermore, construction noise was characterised by sporadic, sudden, and sharp noises, such as hammering, clanging, and drilling.

Like the sudden movements of visitors, these noises could potentially startle, stress and/or introduce uncertainty (Hemsworth, 2003; Rushen et al., 1999) and therefore cause a behavioural response in the animals, in this case a decrease in environmental engagement (defined by reduced foraging and increased retreat behaviour). While construction sites are often seen as dynamic and visually engaging due to the presence of workers and ongoing activities, it is also worth noting that in zoo facilities, or other captive environments, mitigation measures are often put in place to shield animals from the disturbance of construction events. These measures typically involve stationary barriers (hoarding screens) that obscure the direct view of the construction site from the animals. Therefore, these barriers represent the main visual cue perceived by animals and the argument that construction sites are inherently dynamic and visually stimulating overlooks the reality for animals in captive environments.

While specific research on the impact of construction events is limited, spatial avoidance appears to be a common coping mechanism for a variety of animals exposed to such stimuli (Chosy et al., 2014; Jakob-Hoff et al., 2019). For instance, a study at an urban zoo in New Zealand showed that some of the study species i.e., giraffes (*Giraffa camelopardalis*), elephants (*Elephas maximus*), and emus (*Dromaius novaehollandiae*), moved to quieter areas when exposed to continuous or intermittent construction noise (Jakob-Hoff et al., 2019). Studies on callitrichids (common marmoset, *Callithrix jacchus*; cotton-top tamarin, *Saguinus oedipus*; and pied tamarins) have also shown that when given the choice between quiet and noisy areas (featuring music, waterfalls, and white noise), the quiet areas are preferred (McDermott and Hauser, 2007; Wark et al., 2023). In addition, the use of space analysis demonstrated that height played a significant role when tamarins were confronted with either visual or acoustic components individually. Visual and acoustic components for both visitors and construction stimuli caused pied tamarins to retreat to greater heights. This is in line with past research that showed that there is a vertical axis of fear in arboreal primates, meaning that they feel safer with increasing height (Emerson et al., 2011; Nowak et al., 2014). Given that pied tamarins are a small arboreal primate species, it is unsurprising that they seek higher ground when faced with uncertainty.

#### 4.3. Combined stimuli in comparison to stimuli in isolation

Although some components might dominate over others in specific contexts, the combination of information received via multiple sensory modalities can provide a more comprehensive understanding of the environment, potentially improving decision-making (Narins et al., 2005; Partan et al., 2010; Partan and Marler, 2005). Multisensory integration, i.e., the combined use of the senses, appears to be adaptive for behavioural tasks such as foraging, orientation, mate choice, or predator/prey interactions (Higham and Hebets, 2013; Partan et al., 2010). Based on this, we hypothesized those congruous combinations of components would enable pied tamarins to evaluate and understand stimuli more effectively than incongruous stimuli, allowing them to determine whether or not it posed a threat and respond accordingly.

Our results support this hypothesis in the case of visitor stimuli, where congruent combinations, i.e., visitor visual and acoustic stimuli in combination, significantly reduced the vigilance behavioural response of pied tamarins (PC1), compared to components in isolation. One possible explanation for a diminished response may lie in the context presentation. The visual component alone may lack sufficient context, while the addition of acoustic information creates a fuller sensory picture, enabling tamarins to assess the situation more effectively (Hebets and Papaj, 2005; Smith and Belk, 2001). This aligns with findings from studies on other species. For example, Partan et al. (2010) demonstrated that grey squirrels (*Sciurus carolinensis*) responded more strongly to combined alarm signals of a biomimetic squirrel (visual tail flags and auditory alarm barks) than to the presentation of either component in isolation, highlighting the additive value of congruent sensory

information. Similarly, Hazlett and McLay (2005) found that intertidal crabs (*Heterozius rotundifrons*) reduced their freeze response when exposed to combined visual and olfactory cues. Although Hazlett and McLay (2005) suggested that the crabs might switch to a different anti-predatory strategy of finding shelter, the study still concluded that more information is conveyed, and more context is provided when multiple cues were presented through different sensory components.

However, our results reveal a different pattern for construction stimuli, where congruent conditions did not significantly reduce behavioural responses. This difference may be explained by examining the behavioural response to each of these individual components. For the visitor stimulus, both the visual and acoustic components significantly increased vigilance behaviour independently, suggesting that their combination offered consistent and complementary information, enabling tamarins to evaluate the stimulus and moderate their response more effectively. In this case, the combination of both components led to a significant reduction in vigilance behaviour. In contrast, for construction stimuli, only the acoustic component elicited an 'environmental engagement' response (PC2), while the visual component did not lead to any clear behavioural change. This suggests that the visual and acoustic elements of construction stimuli are perceived differently, resulting in nonadditive effects when combined, and instead influence behaviour in different ways. These findings emphasise the importance of considering separate components of stimuli to fully understand animals' behavioural responses. By investigating how animals perceive and respond to isolated and combined components, we can develop more targeted approaches to mitigate and address the impacts of anthropogenic disturbances, both in captivity and in the wild.

#### 4.4. Future research directions

Our results show that behavioural responses in zoo-housed pied tamarins differ when visual and auditory components of stimuli are presented in isolation compared to when they are combined. These results have important husbandry, management, and welfare implications for pied tamarins, other callitrichids, or potentially other small arboreal mammals (Anderson, 2014; Bartlett et al., 2023; Browning and Maple, 2019; Hebert and Bard, 2000; Ross and Lukas, 2006).

##### 4.4.1. Reducing the impact of visitors

Since visitors have been shown to induce strong behavioural changes compared to baseline, especially increased vigilance and proximity to visitors, measures that limit close contact with visitors is recommended to ensure the welfare of these sensitive species, (Sherwen and Hemsworth, 2019; Wormell et al., 1996). More importantly, our results showed that the congruent combination of visual and auditory visitor stimuli reduced behavioural responses across all three PCs, highlighting the importance of allowing pied tamarins access to both sensory modalities to evaluate potential threats.

Given these findings, mitigation measures such as full visual barriers or relocating animals to off-show enclosures should be carefully considered. While these strategies may reduce distress, they could inadvertently limit tamarins' ability to assess and respond appropriately to their environment. Instead of removing the visual aspect of visitors for the animal, allowing the animal to choose and control their visibility of visitors, could offer a good alternative. For example, enclosures with multiple areas, e.g., an outdoor compartment in addition to an indoor compartment, or those that offer partial visibility, e.g., dense vegetation or camouflage netting, allow animals to retreat or engage with their surroundings. Clark et al. (2012) found that privacy screens inside the enclosure and on external windows allowed zoo-housed Western lowland gorillas (*Gorilla gorilla gorilla*) to have more control over visual access to visitors, which reduced their levels of vigilance. Several studies have found that offering choice within enclosures reduces stereotypical or undesired behaviours (Badihi, 2006; Ross, 2006), promotes social behaviours (Kurtycz et al., 2014; Rust et al., 2024), and decreases

physiological stress (Kaplan et al., 2012; Owen et al., 2004), thus positively affecting welfare.

#### 4.4.2. Reducing the impact of construction

Traditional mitigation measures often involve stationary barriers that buffer the sounds but also obscure the direct view of the construction site from the animals. Although sound barriers can effectively dampen noise levels (Orban et al., 2017), the cost of not being able to see the source of noise could potentially outweigh the benefits of reduced noise intensity. Future research should investigate the use of alternative barriers and evaluate the trade-off between visual and sound penetration of such barriers. For example, dense vegetation, such as trees, shrubs, and ground cover, may provide a good alternative. Natural barriers provide partial visibility while also effectively reducing noise levels (Fan et al., 2019; Ozdemir et al., 2014; Yofianti and Usman, 2021). Furthermore, natural barriers are more naturalistic and visually pleasing compared to the usual metal barriers, and thus also enhance the visual aesthetics of the zoo environment (Hong and Jeon, 2014).

In addition, although retreat were the primary responses when pied tamarins were confronted with construction noise (PC2), we recommend not confining the animals to their inside enclosures when such activity is underway. It is important to allow animals choice by giving them permanent access to all parts of the enclosure, especially during construction or similar loud events. A highly enriched and foliage-dense outdoor enclosure design further can also help encourage outside activity to further reduce stress (Kaplan et al., 2012; Sha et al., 2016).

#### 4.4.3. Limitations and implications for captive and wild populations

While this study is one of the first studies attempting to experimentally disentangle visual and acoustic components of different stimuli, it is important to recognise some methodological limitations. Most of these limitations arose from practical considerations of the experimental design. First, visitors were represented by volunteers and thus subtle differences from genuine visitor interactions cannot be ruled out. However, their unfamiliarity and lack of prior contact with the subjects, as well as their visitor-like behaviour and appearance ensured they were likely to be perceived similarly, allowing us to treat them as “visitors” in this context of this experiment. Similarly, the audio recordings were based on simple conversations among volunteers and therefore may not fully capture the variability of real zoo soundscapes. This approach was adopted due to ethical and data protection reasons, as it ensured full consent for recordings and avoided capturing confidential or sensitive information. However, the recordings included natural conservation features, such as spontaneous laughter and brief pauses and were edited to match the acoustic characteristics of typical visitor noise. The construction visual stimulus was also simplified, consisting of two red and white plastic cones and a yellow sign. While a larger installation incorporating construction equipment or machinery (e.g. scaffolding, cement mixer, or power tools), practical constraints limited this choice. In addition, because pied tamarins are a critically endangered and highly sensitive species, all stimuli also had to strike a careful balance between novelty, familiarity and intensity to evoke measurable reactions without causing undue stress. Future studies would benefit from extending this approach to larger sample sizes, across multiple institutions, and species. This would help strengthen the robustness of the results, as well as clarify the broader relevance of multimodal anthropogenic stimuli for animal behaviour. Despite these methodology constraints, this study provided valuable insight into the behavioural responses of captive pied tamarins to isolated and combined visual and acoustic components of anthropogenic stimuli and highlights the importance of accounting for the multimodal nature of stimuli, as well as offering important recommendations for future research.

While captive studies allow for controlled observations, their findings may not fully translate to wild populations. Captive animals, living within the anthropogenic environment and being constrained behind physical barriers, are less able to evade anthropogenic impacts.

Although anthropogenic stimuli, including humans themselves, are stimuli that animals have had little evolutionary time to adapt to, their ‘natural’ behaviour might already be influenced and/or modified by anthropogenic conditions, leading to potentially different responses from those of their wild counterparts. It should also be noted that although pied tamarins in the wild have a better capacity to avoid anthropogenic stimuli, their home range is extremely restricted, making them highly impacted by human pressure. Therefore, further studies are needed to investigate whether effects similar to those we observed also occur in the wild.

## 5. Conclusion

This study emphasised the importance of considering the sensory components of stimuli both individually and in combination. Overall, our results show the diverse responses exhibited by pied tamarins when exposed to different stimuli, with specific components playing a pivotal role in shaping their reactions. To fully understand the ecological impact of human activity and these complex anthropogenic disturbances on animal welfare, future research should account for the fact that stimuli often contain multiple sensory components, which can independently drive behavioural responses. The findings also highlight the importance of considering appropriate mitigation measures to support captive husbandry needs, welfare, and ultimately the conservation of the pied tamarin.

## CRedit authorship contribution statement

**Dunn Jacob:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Fiene Steinbrecher:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Price Eluned:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Clark Fay:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Wascher Claudia:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

## Contributions

Fiene Steinbrecher: study design, data collection, data analysis and drafting/reviewing the manuscript.

Eluned Price: study design, data analysis and revising the manuscript.

Claudia Wascher: study design, data analysis and revising the manuscript.

Fay Clark: study design, data analysis and drafting/reviewing the manuscript

Jacob Dunn: study design, data analysis and drafting/reviewing the manuscript.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

We thank all members of the Mammal Department at Jersey Zoo and student Suha Subhani for their cooperation and help with the study. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.



## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.applanim.2026.106926](https://doi.org/10.1016/j.applanim.2026.106926).

## References

- Altmann, J., 1974. Observational study of behavior: sampling methods. *Behaviour* 49 (3–4), 227–266. <https://doi.org/10.1163/156853974X00534>.
- Anderson, M.R., 2014. Reaching new heights: the effect of an environmentally enhanced outdoor enclosure on gibbons in a zoo setting. *J. Appl. Anim. Welf. Sci.* 17 (3), 216–227. <https://doi.org/10.1080/10888705.2014.916172>.
- Armstrong, D.M., Santymire, R.M., 2013. Hormonal and behavioral variation in pied tamarins housed in different management conditions. *Zoo. Biol.* 32 (3), 299–306. <https://doi.org/10.1002/zoo.21023>.
- Badihi, I. (2006). The effects of complexity, choice and control on the behaviour and the welfare of captive common marmosets (*Callithrix jacchus*) [Doctoral dissertation, University of Stirling].
- Bartlett, A., Grinstead, L., Freeman, M.S., 2023. Behaviour, furnishing and vertical space use of captive callimico (*Callimico goeldii*): implications for welfare. *Animals* 13 (13), 2147. <https://doi.org/10.3390/ANI13132147/S1>.
- Bassett, L., Buchanan-Smith, H.M., McKinley, J., & Smith, T.E. (2003). Effects of Training on Stress-Related Behavior of the Common Marmoset (*Callithrix jacchus*) in Relation to Coping With Routine Husbandry Procedures. [https://doi.org/10.1207/S15327604JAWS0603\\_07](https://doi.org/10.1207/S15327604JAWS0603_07).
- Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models Using lme4. *J. Stat. Softw.* 67 (1). <https://doi.org/10.18637/jss.v067.i01>.
- Bishop, T.D., Brown, J.A., 1992. Threat-sensitive foraging by larval threespine sticklebacks (*Gasterosteus aculeatus*). *Behav. Ecol. Sociobiol.* 31 (2), 133–138. <https://doi.org/10.1007/BF00166346/METRICS>.
- Blumstein, D.T., Patton, M.L., Saltzman, W., 2006. Faecal glucocorticoid metabolites and alarm calling in free-living yellow-bellied marmosets. *Biol. Lett.* 2 (1), 29. <https://doi.org/10.1098/RSBL.2005.0405>.
- Brooks, M.E., Kristensen, K., van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Maechler, M., Bolker, B.M. (2017). Modeling Zero-Inflated Count Data With glmmTMB. *bioRxiv preprint bioRxiv:132753*; doi: (<https://doi.org/10.1101/132753>).
- Browning, H., Maple, T.L., 2019. Developing a metric of usable space for zoo exhibits. *Front. Psychol.* 10 (APR), 449992. <https://doi.org/10.3389/FPSYG.2019.00791/BIBTEX>.
- Budaev, S.V., 2010. Using principal components and factor analysis in animal behaviour research: caveats and guidelines. *Ethology* 116 (5), 472–480. <https://doi.org/10.1111/J.1439-0310.2010.01758.X>.
- Cabana, F., Maguire, R., Hsu, C., Da, Plowman, A., 2018. Identification of possible nutritional and stress risk factors in the development of marmoset wasting syndrome. *Zoo. Biol.* 37 (2), 98–106. <https://doi.org/10.1002/ZOO.21398>.
- Chiew, S.J., Butler, K.L., Sherwen, S.L., Coleman, G.J., Fanson, K.V., Hemsworth, P.H., 2019. Effects of regulating visitor viewing proximity and the intensity of visitor behaviour on little penguin (*eudyptula minor*) behaviour and welfare. *Animals* 2019 9 (6), 285. <https://doi.org/10.3390/ANI9060285>.
- Chosy, J., Wilson, M., Santymire, R., 2014. Behavioral and physiological responses in felids to exhibit construction. *Zoo. Biol.* 33, 267–274. <https://doi.org/10.1002/zoo.21142>.
- Clark, F.E., Dunn, J.C., 2022. From soundwave to soundscape: a guide to acoustic research in captive animal environments. *Front. Vet. Sci.* 9, 889117. <https://doi.org/10.3389/FVETS.2022.889117/BIBTEX>.
- Clark, F.E., Fitzpatrick, M., Hartley, A., King, A.J., Lee, T., Routh, A., George, K., 2012. Relationship between behavior, adrenal activity, and environment in zoo-housed western lowland gorillas (*Gorilla gorilla*). *Zoo. Biol.* 31 (3), 306–321. <https://doi.org/10.1002/zoo.20396>.
- Collins, C., Corkery, I., Haigh, A., McKeown, S., Quirke, T., O'Riordan, R., 2017. The effects of environmental and visitor variables on the behavior of free-ranging ring-tailed lemurs (*Lemur catta*) in captivity. *Zoo. Biol.* 36 (4), 250–260.
- Costello, A.B., Osborne, J., 2019. Best practices in exploratory factor analysis: four recommendations for getting the most from your analysis. *Pract. Assess. Res. Eval.* 10 (1), 7.
- Davey, G., 2007. Visitors' effects on the welfare of animals in the zoo: a review, 10 (2), 169–183. <https://doi.org/10.1080/10888700701313595>.
- Emerson, S.E., Brown, J.S., Linden, J.D., 2011. Identifying Sykes' monkeys', *Cercopithecus albogularis erythrarchus*, axes of fear through patch use. *Anim. Behav.* 81 (2), 455–462. <https://doi.org/10.1016/J.ANBEHAV.2010.11.018>.
- Fan, J., Hao, H., Fan, R., 2019. Noise reduction effect of plant landscape in city zoo based on animal protection. *Rev. Cient. Ifica De la Fac. De. Cienc. Vet.* 29 (5), 1180. (<https://link.gale.com/apps/doc/A624690440/IFME?u=anon~2fa58ee3&sid=googleScholar&id=14494317>) (+).
- Farrand, A. (2007). The effect of zoo visitors on the behaviour and welfare of zoo mammals. (<http://www.storre.stir.ac.uk/handle/1893/300>).
- Fernandez, E.J., Tamborski, M.A., Pickens, S.R., Timberlake, W., 2009. Animal–visitor interactions in the modern zoo: conflicts and interventions. *Appl. Anim. Behav. Sci.* 120 (1–2), 1–8.
- Ferrari, S.F., 2009. Predation Risk and Antipredator Strategies BT. In: Garber, P.A., Estrada, A., Bicca-Marques, J.C., Heymann, E.W., Strier, K.B. (Eds.), *South American Primates: Comparative Perspectives in the Study of Behavior, Ecology, and Conservation*. Springer New York, pp. 251–277. [https://doi.org/10.1007/978-0-387-78705-3\\_10](https://doi.org/10.1007/978-0-387-78705-3_10).
- Ferrari, S.F., Ferrari, M.A.L., 1990. Predator avoidance behaviour in the buffy-headed marmoset, *Callithrix flaviceps*. *Primates* 31 (3), 323–338. <https://doi.org/10.1007/BF02381104/METRICS>.
- Gordo, M., Callea, F.O., Vasconcelos, S.A., Leite, J.J.F., Ferrari, S.F., 2013. The challenges of survival in a concrete jungle: conservation of the pied tamarin (*Saguinus bicolor*) in the urban landscape of Manaus, Brazil. *Primates* *Fragm. Complex. Resil.* 357–370. [https://doi.org/10.1007/978-1-4614-8839-2\\_23/COVER](https://doi.org/10.1007/978-1-4614-8839-2_23/COVER).
- Gordo, M., Röhe, F., Vidal, M.D., Subirá, R., Boubli, J.P., Mittermeier, R.A., & Jerusalinsky, L. (2021). *Saguinus bicolor* (amended version of 2019 assessment). The IUCN Red List of Threatened Species 2021: e.T40644A192551696. (<https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T40644A192551696.en>). Accessed on 02 July 2024.
- Halfwerk, W., Slabbekoorn, H., 2015. Pollution going multimodal: the complex impact of the human-altered sensory environment on animal perception and performance. *Biol. Lett.* 11 (4). <https://doi.org/10.1098/RSBL.2014.1051>.
- Harding, H.R., Gordon, T.A.C., Eastcott, E., Simpson, S.D., Radford, A.N., 2019. Causes and consequences of intraspecific variation in animal responses to anthropogenic noise. *Behav. Ecol.* 30 (6), 1501–1511. <https://doi.org/10.1093/beheco/arz114>.
- Hart, D., 2007. Predation on primates: a biogeographical analysis. *Primate Anti Predat. Strateg.* 27–59. [https://doi.org/10.1007/978-0-387-34810-0\\_2](https://doi.org/10.1007/978-0-387-34810-0_2).
- Hartig, F. (2022). DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models R package version 0.4.6. (<https://CRAN.R-project.org/package=DHARMa>).
- Hazlett, B.A., McLay, C., 2005. Responses to predation risk: alternative strategies in the crab *Heterosius rotundifrons*. *Anim. Behav.* 69 (4), 967–972. <https://doi.org/10.1016/J.ANBEHAV.2004.06.028>.
- Hebert, P.L., Bard, K., 2000. Orangutan use of vertical space in an innovative habitat. *Zoo. Biol.* 19, 239–251.
- Hebets, E.A., Papaj, D.R., 2005. Complex signal function: developing a framework of testable hypotheses. *Behav. Ecol. Sociobiol.* 57 (3), 197–214. <https://doi.org/10.1007/S00265-004-0865-7/FIGURES/2>.
- Hemsworth, P.H., 2003. Human–animal interactions in livestock production. *Appl. Anim. Behav. Sci.* 81 (3), 185–198. [https://doi.org/10.1016/S0168-1591\(02\)00280-0](https://doi.org/10.1016/S0168-1591(02)00280-0).
- Higham, J.P., Hebets, E.A., 2013. An introduction to multimodal communication. *Behav. Ecol. Sociobiol.* 67 (9), 1381–1388. <https://doi.org/10.1007/S00265-013-1590-X/METRICS>.
- Hong, J.Y., Jeon, J.Y., 2014. The effects of audio–visual factors on perceptions of environmental noise barrier performance. *Landsc. Urban Plan.* 125, 28–37. <https://doi.org/10.1016/J.LANDURBPLAN.2014.02.001>.
- Hosey, G.R., 2000. Zoo animals and their human audiences: what is the visitor effect? *Anim. Welf.* 9 (4), 343–357. <https://doi.org/10.1017/S0962728600022946>.
- Hosey, G., 2013. Hediger revisited: how do zoo animals see us? *J. Appl. Anim. Welf. Sci.* 16 (4), 338–359. <https://doi.org/10.1080/10888705.2013.827916>.
- Jakob-Hoff, R., Kingan, M., Fenemore, C., Schmid, G., Cockrem, J.F., Crackle, A., Bommel, E.Van, Connor, R., Descovich, K., 2019. Potential impact of construction noise on selected zoo animals. *Vol. 9, 504 Animals* 2019 9 (8), 504. <https://doi.org/10.3390/ANI9080504>.
- Kaplan, G., Pines, M.K., Rogers, L.J., 2012. Stress and stress reduction in common marmosets. *Appl. Anim. Behav. Sci.* 137 (3–4), 175–182. <https://doi.org/10.1016/J.APPLANIM.2011.04.011>.
- Kim, Y., Vlaeyen, J.M.R., Heesen, R., Clay, Z., Kret, M.E., 2022. The association between the bared-teeth display and social dominance in captive chimpanzees (pan troglodytes). *Affect. Sci.* 3 (4), 749–760. <https://doi.org/10.1007/S42761-022-00138-1/FIGURES/1>.
- Kurtycz, L.M., Wagner, K.E., Ross, S.R., 2014. The choice to access outdoor areas affects the behavior of great apes. *J. Appl. Anim. Welf. Sci.* 17 (3), 185–197. <https://doi.org/10.1080/10888705.2014.896213>.
- Lê, S., Josse, J., Husson, F., 2008. FactoMineR: a package for multivariate analysis. *J. Stat. Softw.* 25 (1), 1–18. <https://doi.org/10.18637/jss.v025.i01>.
- Lind, J., Cresswell, W., 2005. Determining the fitness consequences of antipredation behavior. *Behav. Ecol.* 16 (5), 945–956. <https://doi.org/10.1093/BEHECO/ARI075>.
- López, J., Wormell, D., Rodríguez, A., 2001. Preliminary evaluation of the efficacy and safety of a UVB lamp used to prevent metabolic bone disease in pied tamarins *Saguinus bicolor* at Jersey Zoo. *Dodo* 37, 41–49. (<https://www.scopus.com/inward/record.uri?eid=2-s2.0-0035566326&partnerID=40&md5=3bc3099f801cc8fd408e69dd38fd4b0a>).
- McDermott, J., Hauser, M.D., 2007. Nonhuman primates prefer slow tempos but dislike music overall. *Cognition* 104 (3), 654–668. <https://doi.org/10.1016/J.COGNITION.2006.07.011>.
- Mittermeier, R.A., Reuter, K.E., Rylands, A.B., Ang, A., Jerusalinsky, L., Nash, S.D., Schwitzer, C., Ratsimbazafy, J. and Humle, T. (2024). *Primates in Peril: The World's 25 Most Endangered Primates 2023–2025*. IUCN SSC Primate Specialist Group, International Primatological Society, Re:wild, Washington, DC. 155pp.
- Monclús, R., Palomares, F., Tablado, Z., Martínez-Fontúrbel, A., Palme, R., 2009. Testing the threat-sensitive predator avoidance hypothesis: physiological responses and predator pressure in wild rabbits. *Oecologia* 158 (4), 615–623. <https://doi.org/10.1007/S00442-008-1201-0/TABLES/2>.
- Morton, F.B., Altschuld, D., 2019. Data reduction analyses of animal behaviour: avoiding Kaiser's criterion and adopting more robust automated methods. *Anim. Behav.* 149, 89–95. <https://doi.org/10.1016/J.ANBEHAV.2019.01.003>.
- Munoz, N.E., Blumstein, D.T., 2012. Multisensory perception in uncertain environments. *Behav. Ecol.* 23 (3), 457–462. <https://doi.org/10.1093/BEHECO/ARR220>.

- Narins, P.M., Grabul, D.S., Soma, K.K., Gaucher, P., Hödl, W., 2005. Cross-modal integration in a dart-poison frog. *Proc. Natl. Acad. Sci.* 102 (7), 2425–2429. <https://doi.org/10.1073/PNAS.0406407102>.
- Neitzel, R., Seixas, N.S., Camp, J., Yost, M., 1999. An assessment of occupational noise exposures in four construction trades. *Am. Ind. Hyg. Assoc. J.* 60 (6), 807–817. <https://doi.org/10.1080/00028899908984506>.
- Nowak, K., Le Roux, A., Richards, S.A., Scheijen, C.P.J., Hill, R.A., 2014. Human observers impact habituated samango monkeys' perceived landscape of fear. *Behav. Ecol.* 25 (5), 1199–1204. <https://doi.org/10.1093/BEHECO/ARU110>.
- Orban, D.A., Soltis, J., Perkins, L., Mellen, J.D., 2017. Sound at the zoo: Using animal monitoring, sound measurement, and noise reduction in zoo animal management. *Zoo. Biol.* 36 (3), 231–236. <https://doi.org/10.1002/zoo.21366>.
- Owen, M.A., Swaisgood, R.R., Czekala, N.M., Steinman, K., Lindburg, D.G., 2004. Monitoring stress in captive giant pandas (*Ailuropoda melanoleuca*): behavioral and hormonal responses to ambient noise. *Zoo. Biol.* 23 (2), 147–164. <https://doi.org/10.1002/ZOO.10124>.
- Ozdemir, B., Bayramoglu, E., Demirel, O., 2014. Noise pollution and human health in trabzon parks. *Stud. EthnoMed.* 8 (2), 127–134. <https://doi.org/10.1080/09735070.2014.11917627>.
- Papworth, S., Milner-Gulland, E.J., Slocombe, K., 2013. Hunted woolly monkeys (*lagotrix poeppigii*) show threat-sensitive responses to human presence. *PLoS ONE* 8 (4), 62000. <https://doi.org/10.1371/journal.pone.0062000>.
- Partan, S.R., Fulmer, A.G., Gounard, M.A.M., Redmond, J.E., 2010. Multimodal alarm behavior in urban and rural gray squirrels studied by means of observation and a mechanical robot. *Curr. Zool.* 56 (3), 313–326. <https://doi.org/10.1093/CZOOLO/56.3.313>.
- Partan, S.R., Marler, P., 2005. Issues in the classification of multimodal communication signals. *Am. Nat.* 166 (2), 231–245. <https://doi.org/10.1086/431246>.
- Petersen, R.M., Dubuc, C., Higham, J.P., 2018. Facial displays of dominance in non-human primates. *Facial Disp. Lead.* 123–143. [https://doi.org/10.1007/978-3-319-94535-4\\_6/COVER](https://doi.org/10.1007/978-3-319-94535-4_6/COVER).
- Price, E., Coleman, R., Ahsmann, J., Glendewar, G., Hunt, J., Smith, T., Wormell, D., 2019. Individual, social, and environmental factors affecting salivary and fecal cortisol levels in captive pied tamarins (*Saguinus bicolor*). *Am. J. Primatol.* 81 (8), e23033.
- R Core Team (2022). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. (<https://www.R-project.org/>).
- Revelle, W. (2023). *\_psych: Procedures for Psychological, Psychometric, and Personality Research*. Northwestern University, Evanston.
- Richards-Rios, P., Wigley, P., López, J., Wormell, D., Barbón, A., 2021. Changes in the faecal microbiome of pied tamarins (*Saguinus bicolor*) associated with chronic, recurrent diarrhoea and weight loss. *Anim. Micro* 3 (1), 1. <https://doi.org/10.1186/s42523-020-00062-4>.
- Ross, S.R., Lukas, K.E., 2006. Use of space in a non-naturalistic environment by chimpanzees (*Pan troglodytes*) and lowland gorillas (*Gorilla gorilla gorilla*). *Appl. Anim. Behav. Sci.* 96 (1–2), 143–152. <https://doi.org/10.1016/J.APPLANIM.2005.06.005>.
- Rushen, J., Taylor, A.A., De Passillé, A.M., 1999. Domestic animals' fear of humans and its effect on their welfare. *Appl. Anim. Behav. Sci.* 65 (3), 285–303. [https://doi.org/10.1016/S0168-1591\(99\)00089-1](https://doi.org/10.1016/S0168-1591(99)00089-1).
- Rust, K., Clegg, I., Fernandez, E.J., 2024. The voice of choice: a scoping review of choice-based animal welfare studies. *Appl. Anim. Behav. Sci.*, 106270 <https://doi.org/10.1016/j.applanim.2024.106270>.
- Sha, J.C.M., Ismail, R., Marlena, D., Lee, J.L., 2016. Environmental complexity and feeding enrichment can mitigate effects of space constraints in captive callitrichids. *Lab. Anim.* 50 (2), 137–144. <https://doi.org/10.1177/0023677215589258/ASSET/IMAGES/LARGE/10.1177.0023677215589258-FIG4.JPEG>.
- Shannon, G., McKenna, M.F., Angeloni, L.M., Crooks, K.R., Fristrup, K.M., Brown, E., Warner, K.A., Nelson, M.D., White, C., Briggs, J., McFarland, S., Wittemyer, G., 2016. A synthesis of two decades of research documenting the effects of noise on wildlife. *Biol. Rev.* 91 (4), 982–1005. <https://doi.org/10.1111/brv.12207>.
- Sherwen, S.L., Hemsworth, P.H., 2019. The visitor effect on zoo animals: implications and opportunities for zoo animal welfare. *Animals* 9 (6), 366. <https://doi.org/10.3390/ani9060366>.
- Smith, J.J., 2014. Human–animal relationships in zoo-housed orangutans (*P. abelii*) and gorillas (*G. g. gorilla*): the effects of familiarity. *Am. J. Primatol.* 76 (10), 942–955. <https://doi.org/10.1002/AJP.22280>.
- Smith, J.J. (2016). Primates and People in the Zoo: Implications of Human–Animal Interactions and Relationships. 371–398. ([https://doi.org/10.1007/978-3-319-30469-4\\_20](https://doi.org/10.1007/978-3-319-30469-4_20)).
- Smith, M.E., Belk, M.C., 2001. Risk assessment in western mosquitofish (*Gambusia affinis*): do multiple cues have additive effects? *Behav. Ecol. Sociobiol.* 51 (1), 101–107. <https://doi.org/10.1007/S002650100415/METRICS>.
- Sobroza, T.V., Cerqueda, L.S., Simões, P.I., Gordo, M., 2017. Vocal repertoire and its behavioral contexts in the pied tamarin, *Saguinus bicolor*. *Int. J. Primatol.* 38 (4), 642–655. <https://doi.org/10.1007/s10764-017-9971-z>.
- Sobroza, T.V., Dunn, J.C., Gordo, M., Barnett, A.A., 2024a. Do pied tamarins increase scent-marking in response to urban noise? *Ethol. Ecol. Evol.* 36 (2), 136–149. <https://doi.org/10.1080/03949370.2023.2248591>.
- Sobroza, T.V., Gordo, M., Dunn, J.C., Pequeno, P.A.C.L., Naissinger, B.M., Barnett, A.P. A., 2024b. Pied tamarins change their vocal behavior in response to noise levels in the largest city in the Amazon. *Am. J. Primatol.* 86 (5), e2360. <https://doi.org/10.1002/ajp.23606>.
- Steinbrecher, F., Dunn, J.C., Price, E.C., Buck, L.H., Wascher, C.A., Clark, F.E., 2023. The effect of anthropogenic noise on foraging and vigilance in zoo housed pied tamarins. *Appl. Anim. Behav. Sci.* 265, 105989.
- Suárez, P., Recuerda, P., Arias-de-Reyna, L., 2017. Behaviour and welfare: the visitor effect in captive felids. *Anim. Welf.* 26 (1), 25–34.
- Wale, M.A., Simpson, S.D., Radford, A.N., 2013. Noise negatively affects foraging and antipredator behaviour in shore crabs. *Anim. Behav.* 86 (1), 111–118. <https://doi.org/10.1016/j.anbehav.2013.05.001>.
- Wark, J.D., Schook, M.W., Dennis, P.M., Lukas, K.E., 2023. Do zoo animals use off-exhibit areas to avoid noise? A case study exploring the influence of sound on the behavior, physiology, and space use of two pied tamarins (*Saguinus bicolor*). *Am. J. Primatol.* 85 (3), e23421. <https://doi.org/10.1002/ajp.23421>.
- Wormell, D., Brayshaw, M., Price, E., Herron, S., 1996. Pied tamarins *Saguinus bicolor* at the Jersey Wildlife Preservation Trust: management, behaviour and reproduction. *Dodo* 32 (0), 76–97.
- Yofianti, D., Usman, K., 2021. Relationship of plant types to noise pollution absorption level to improve the quality of the road environment. *IOP Conference Series Earth Environmental Science* 926 (1), 012074. <https://doi.org/10.1088/1755-1315/926/1/012074>.